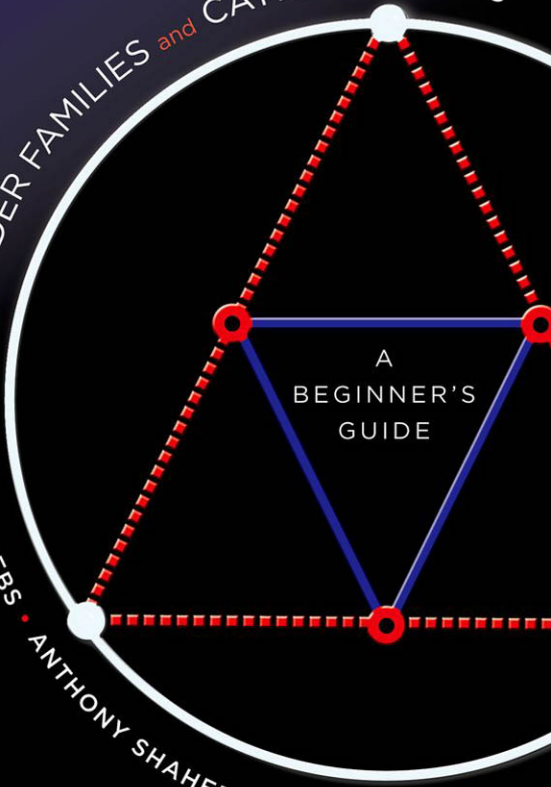
EXPANDER FAMILIES and CAYLEY GRAPHS

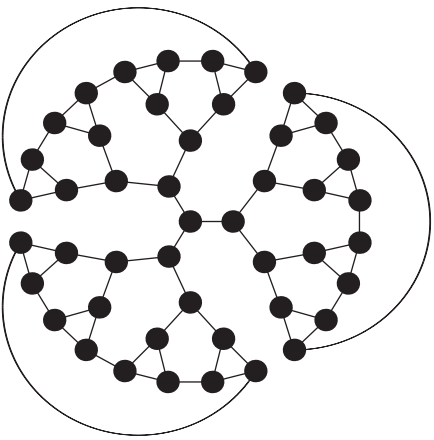


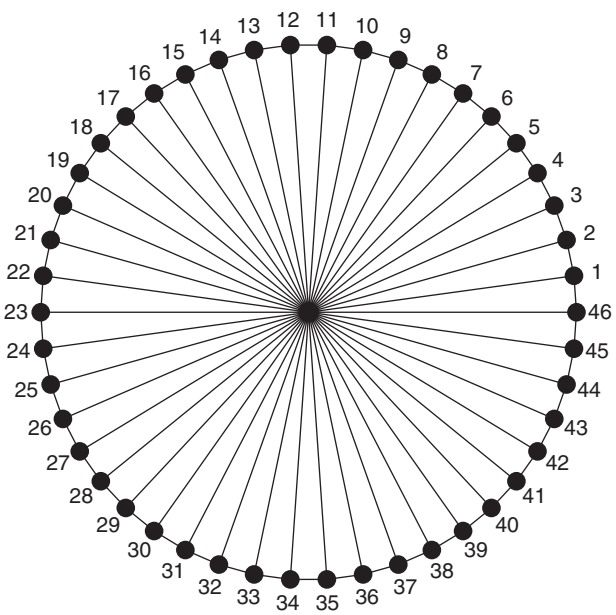
A BEGINNER'S GUIDE

MIKE KREBS • ANTHONY SHAHEEN

Expander Families and Cayley Graphs

Consider the two graphs here. Each has 46 vertices, and each has 3 edges per vertex. Which one would make a better communications network, and why?





See the Introduction for more details and a discussion of how this question leads us naturally into the fascinating theory of *expander families*.

Expander Families and Cayley Graphs

A Beginner's Guide

MIKE KREBS

AND

ANTHONY SHAHEEN

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PREFACE

ABOUT THIS BOOK

This book provides an introduction to the mathematical theory of expander families. It is intended for advanced undergraduates, graduate students, and faculty.

The prerequisites for this book are as follows. No graph theory is assumed; we develop it all from scratch. One course on introductory undergraduate group theory is assumed. One course on linear algebra is assumed. We provide Appendix A as a refresher on linear algebra for those who need it. In particular, those who are reading the book should either know the spectral theorem (Theorem A.53) or be willing to take it for granted. With these parameters, the book is self-contained. Analysis is helpful but not necessary; we occasionally encounter statements of the form "For all $\epsilon > 0$, there exists N such that if $n \geq N$, then ..." Appendix B gives a review of big oh notation and the limit inferior of a sequence. This appendix uses the definition of the limit of a sequence. At one point in Chapter 8, we use the fact that a continuous function obtains its maximum on a compact subset of $\mathbb{C}^n$. In Section 6 of Chapter 7, we use the fact that the multiplicative subgroup of the set of integers modulo a prime is cyclic.

Figure P.1 gives a flow chart for the book. For example, Chapter 1 is a prerequisite for every other chapter. To read Chapter 7 one must first read Chapters 1 and 6.

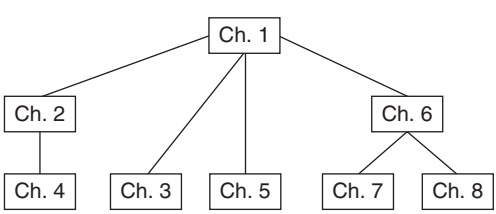


Figure P.1

We make several points with regard to Figure P.1.

1. Two concepts briefly introduced in Chapter 3 are the definition of Ramanujan graph and the notation $\lambda(X)$, both of which are used often in later chapters. We have not indicated this dependence in Figure P.1.

When readers encounter these objects, they can quickly read the definition in Chapter 3—there is no need to read all of Chapter 3.

2. The sections on subgroups and quotients in Chapters 4 and 8 make use of concepts and notations from Chapter 2 (especially the section on Schreier generators).
3. Chapter 5 may be read directly after reading Chapter 1. However, if one wants to read Section 5 of Chapter 5, one must learn the definition of the semidirect product. This definition is given in Section 6 of Chapter 4.
4. Chapter 7 uses only Section 1–4 of Chapter 6. Section 6 of Chapter 6 is used only in Section 4 of Chapter 8.

Each chapter of the book contains a Notes section, where we provide additional information on expander families and related material. We also give references to the literature. The Notes sections are extensive, and we encourage the reader to browse through them.

Many chapters end with a list of student research problems. These problems are intended for independent studies, Research Experience for Undergraduates programs (REUs), and research projects for theses or capstone projects. The problems can be attacked by advanced undergraduates, graduate students, and faculty interested in the area.

The authors have used this book for several courses. In a 10-week course for advanced undergraduates and master's students we covered the following: all of Chapter 1; Section 1 and Section 2 of Chapter 3; Section 1, Section 2, and Section 3 of Chapter 4; and all of Chapter 5. Shaheen used the textbook in a 10-week undergraduate capstone course. In that course, he covered some of Chapter 1. Shaheen used the course for several undergraduate and independent studies courses. In one course he covered Chapter 1 and Chapter 3; in another he covered Chapter 6; in another he covered Chapter 6 and Chapter 8.

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NOTATIONS AND CONVENTIONS

If V is a set, we write $S \subset V$ to indicate that S is a (not necessarily proper) subset of V.

If S and V are multisets, then the notation $S \subset V$ will indicate that the multiplicity of any element of S is less than or equal to its multiplicity as an element of V.

Let A and B be sets. We write $A \setminus B$ for set difference; that is $A \setminus B = \{x \in A \mid x \notin B\}$.

If X, Y, and Z are sets with $X \subset Y$ and $f : Y \to Z$, then we use the notation $f|_X$ to denote the restriction of f to X.

If G is a group, we write $H < G$ to indicate that H is a subgroup of G and $H \lhd G$ to indicate that H is a normal subgroup of G.

We write $G = 1$ to indicate that G is the trivial group.

We denote the group of integers modulo n under addition by $\mathbb{Z}_n$.

We denote the sets of integers, real numbers, and complex numbers, respectively, by $\mathbb{Z}$, $\mathbb{R}$, and $\mathbb{C}$.

Let x be a real number. We write $\lfloor x \rfloor$ for the greatest integer less than or equal to x. For example, $\lfloor 1.64 \rfloor = 1$.

We sometimes write $\exp(z)$ for e^z.

If $\sigma, \tau \in S_n$, then we multiply as follows: $\sigma\tau = \sigma \circ \tau$. We write elements of S_n in cycle notation, using commas to seperate the entries. For example, $(1, 2)(1, 3) = (1, 3, 2)$.

The dihedral group is given by

$$D_n = \{1, r, r^2, \ldots, r^{n-1}, s, sr, \ldots, sr^{n-1}\}$$

where $r^n = 1, s^2 = 1$, and $rs = sr^{-1}$.

We use ^{t} to denote the transpose of a matrix. To save space, we write column vectors as the transpose of a row vector; that is,

$$(x_1, x_2, \ldots, x_n)^t = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

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INTRODUCTION

1. WHAT IS AN EXPANDER FAMILY?

A graph is a collection of dots (*vertices*, singular *vertex*) and connections (*edges*) between them. (We make this definition precise in Chapter 1; for now, our discussion will be informal and intuitive.) Figure I.1 shows a graph Y, and Figure I.2 shows a graph Z.

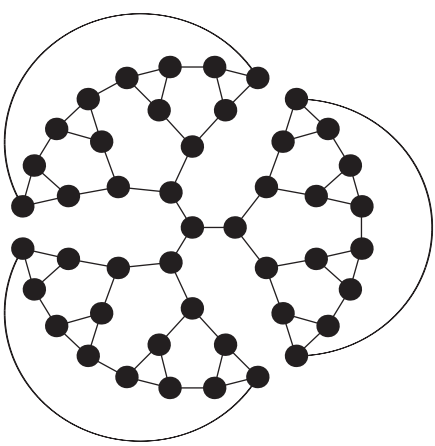


Figure I.1 A 3-regular graph Y with 46 vertices

We introduce a way of thinking about graphs that will help us develop many of the mathematical concepts in this book. Regard a graph as a communications network. The vertices represent entities we wish to have communicate with one another, such as computers, telephones, or small children holding tin cans. The edges represent connections between them, such as fiber optic cables, telephone lines, or pieces of string tied to the cans. Two vertices can communicate directly with one another iff there is an edge that runs between them (i.e., if the two vertices are *adjacent*). Communication is instantaneous across edges, but there may be delays at the vertices (because that's where the humans are). The distance

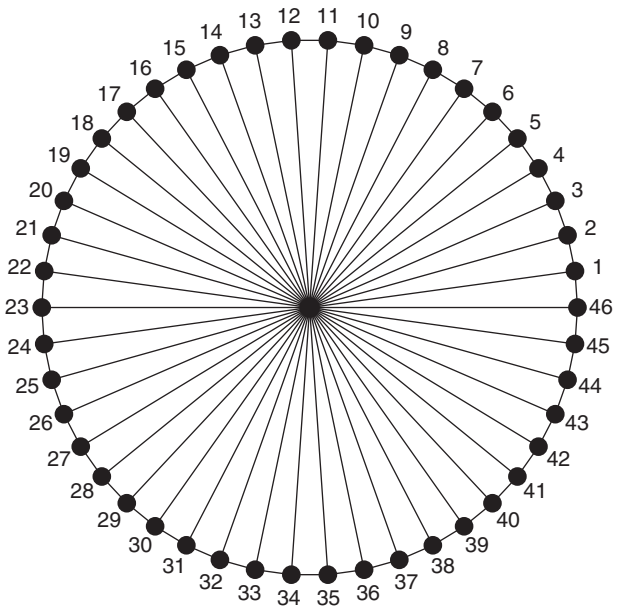


Figure I.2 A 3-regular graph Z with 46 vertices

between two vertices is irrelevant. Likewise, whether two edges appear to cross is irrelevant. (We can always run one wire above another, putting some space between them.) We would like to have a large number of vertices, so that many people can communicate with one another. However, cables are expensive, so we want to get by with as few edges as possible.

Suppose we are told to design a communications network with 46 vertices such that each vertex is adjacent to exactly 3 other vertices (in which case, we say that the graph is 3-*regular*). One way to do this would be as in the graph Y in Figure I.1. An alternative is to use the graph Z in Figure I.2. (Note that in the graph Z, there is no vertex in the center; this is simply an illusion caused by all the edges that appear to cross there. Each vertex in Z is adjacent to the two next to it as well as the one directly across from it. Vertex 3, for example, is adjacent to vertices 2, 4, and 26.)

Of the graphs Y and Z, which one is a better communications network, and why?

(We realize that this question is not precise, for we have not told you what "better" means. Indeed, the point here is for *you* to think about what constitutes a "better" communications network. Enjoy this opportunity to take a stab at a math question that truly has no right or wrong answer!)

We could make a strong case for Y by arguing that it is a *faster* network than Z. In Y, we can get from any vertex to any other vertex in no more than eight steps, by first going to the center of the graph and then back outward. To get from vertex 1 to vertex 12 in Z, however, requires at least 11 steps.

On the other hand, one could argue equally well that Z is the better communications network, for it is *more reliable* than Y. If we cut just one of the edges in Y attached to the center vertex, then the graph becomes disconnected: 15 people will have no way to communicate with the other 31. In contrast, you can cut any edge in Z, and the graph will remain connected, that is, any two vertices can still send messages to one another. (In fact, cutting any *two* edges in Z still leaves behind a connected graph.)

Perhaps to the reader's dismay, we leave unresolved the question of whether Y or Z is the superior communications network. Discuss the matter at your own peril; we take no responsibility for any barroom brawls that may ensue. Our intention in asking this question was merely to find some natural ways of measuring a graph's quality as a network, and now we have two such ways, namely, its speed (as measured by the minimum number of edges required, even in the worst case, to get from one vertex to another, i.e., the graph's *diameter*) and its reliability (as measured by the minimum number of edge cuts needed to disconnect the graph, i.e., its *edge connectivity*).

As it turns out, a single quantity gives us information about both the speed and the reliability of a communications network. Let's probe the graphs Y and Z a bit more deeply to see a sense in which we can view diameter and edge connectivity as two sides of the same coin.

Consider the set V_n of vertices we can get to in no more than n steps from a fixed base vertex. If a communications network is to be fast—that is, if the graph is to have small diameter—then we expect to have many edges from V_n to its complement. In that way, the sets V_n grow rapidly as n increases; in other words, we can reach many points in relatively few steps. (One technicality: we must restrict our attention to sets V_n which contain no more than half of the graph's vertices, or else we will start running out of vertices in the complement.)

Let's see what goes wrong in our "slow" graph Z. Consider the set S of vertices we can reach in no more than four steps, starting at vertex 1. Then $S = \{1, 2, 3, 4, 5, 21, 22, 23, 24, 25, 26, 27, 43, 44, 45, 46\}$, as shown by the white vertices in Figure I.3. Note that while S contains sixteen vertices, there are only six edges going from S to its complement. (These are the six dashed edges in Figure I.3. They connect 5 and 28, 20 and 43, 5 and 6, 20 to 21, 27 to 28, and 42 and 43.) The graph Z would make a slow network because it contains this set S with many vertices but relatively few edges from S to its complement.

Now, what makes Y unreliable? Let S be the set of 15 white vertices in Figure I.4. Then there is only one edge going from S to its complement. (It's the dotted edge in Figure I.4.) Cutting this edge disconnects those 15 vertices from the rest of the graph. The graph Y would make an unreliable network because it contains this set S with many vertices but relatively few edges from S to its complement.

Our discussion suggests a graph invariant that we may wish to measure. For the reason cited earlier, we restrict ourselves to those sets containing no more than half of the graph's vertices. For all such sets S, we take the minimum ratio of the number of edges between S and its complement to the size of the set S. This invariant is called the *isoperimetric constant* of the graph X and is denoted $h(X)$.

Let's now consider an infinite family of 2-regular graphs: the cycle graphs C_n, some of which are depicted in Figure I.5.

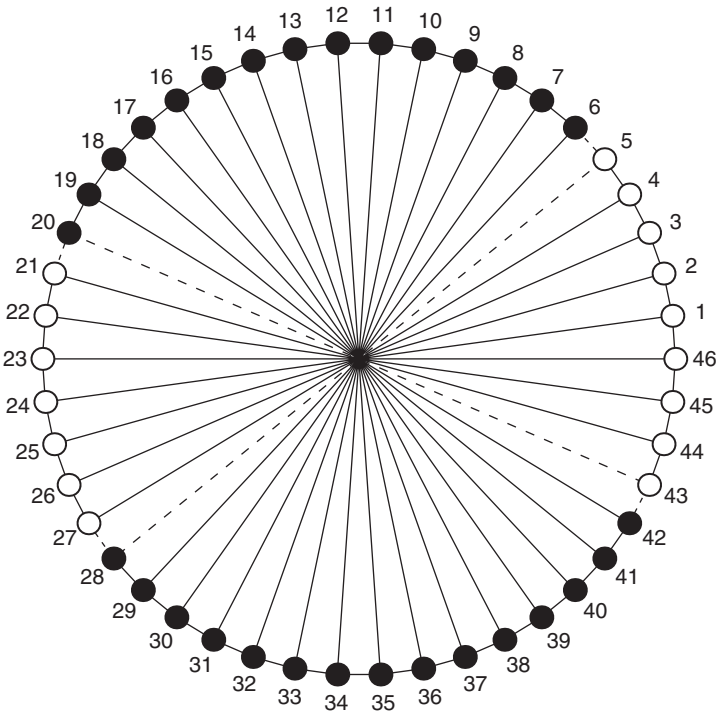


Figure I.3 Graph Z and set S with dotted lines for boundary

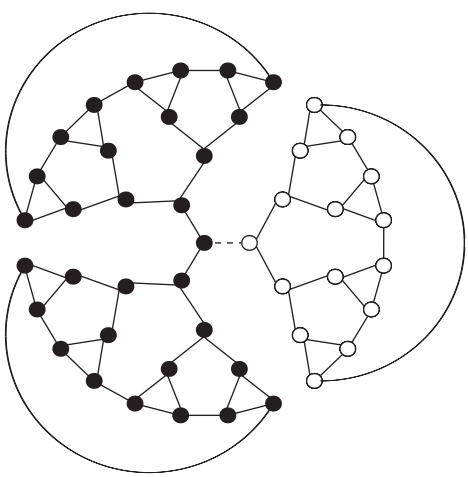


Figure I.4 Graph Y and set S with dotted lines for boundary

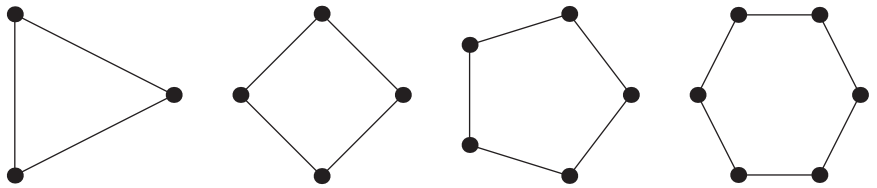


Figure I.5 The cycle graphs C_3, C_4, C_5, and C_6

Take the set S to be the "bottom half" of vertices in C_n. For example, the white vertices in Figure I.6 give S for C_8. (We assume for now that n is even.) Then S contains $n/2$ vertices, and there are two edges going from S to its complement. So we get the ratio $2/(n/2) = 4/n$. This ratio may or may not be the minimum, so $h(C_n) \leq 4/n$ for n even. A similar argument shows that $h(C_n) \leq 4/(n-1)$ for n odd. Hence we see that as $n \to \infty$, the isoperimetric constant of the cycle graph C_n goes to 0. This fact matches well with our intuition that the quality of the cycle graphs as communications networks becomes poorer as they become larger.

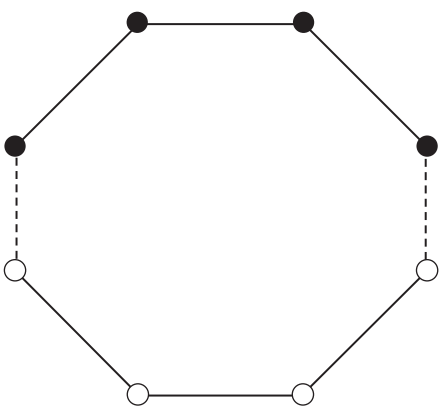


Figure I.6 C_8 with bottom half chosen and boundary dotted

We may be tempted to conjecture, based on this example, that if d is a fixed natural number and (X_k) is a sequence of d-regular finite graphs, where the number of vertices goes to infinity, then $h(X_k) \to 0$. Surprisingly, this statement is false. In 1973, Pinsker [108] used a probabalistic argument to demonstrate that for any integer $d \geq 3$, there exists $\epsilon > 0$ and an unbounded sequence (X_k) of d-regular graphs such that $h(X_k) \geq \epsilon$ for all k. Such a sequence (X_k) is called an *expander family*.

Pinsker's existence proof was non-constructive. In 1973, Margulis [92] used high-powered algebraic techniques to give the first explicit construction of an expander family. In the early twenty-first century, mathematicians began employing more elementary combinatorial methods to explicitly construct expander families

(we discuss one of these methods in Chapter 5). The purpose of this book is to give a brief and accessible introduction to the theory of expander families, particularly in light of recent developments in combinatorial techniques.

It turns out that if one chooses a sequence of d-regular graphs "at random," it is almost certain to be an expander family. (See the Notes sections at the end of Chapters 1 and 3 for some precise statements in this vein.) Nevertheless, explicitly constructing an expander family is nontrivial. The situation is not unlike that of transcendental numbers. If one chooses a real number "at random," it is almost certain to be transcendental. However, it is by no means easy to prove that any particular number is transcendental.

2. WHAT IS A CAYLEY GRAPH?

Before attempting to construct expander families, we first tackle a much more basic question: how does one produce an unbounded sequence of d-regular graphs at all?

We have seen one such construction already, namely, the cycle graphs C_n. Label the vertices of C_n with the elements of the group $\mathbb{Z}_n$ of integers modulo n, as in Figure I.7. The vertex a is adjacent to the vertices $a + 1$ and $a - 1$. We might say that two ingredients give us this graph, namely, the group $\mathbb{Z}_n$ and the set of "generators" 1 and -1. Moreover, the graph is 2-regular because we have chosen two generators.

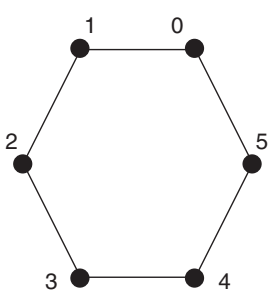


Figure I.7 Cycle graph C_6

The same two ingredients produce the graph Z in Figure I.2. Here, the group is $\mathbb{Z}_{46}$ and the generators are 1, -1, and 23. There are three generators, hence Z is 3-regular.

The general recipe is the classical Cayley graph construction. We start by taking a group G and some subset Γ of G. The elements of G become the vertices of the graph. We draw an edge from x to $x\gamma$ for every $\gamma \in \Gamma$. (Some care must be taken to ensure that the graph is undirected—see Section 1.2 for all the gory details.)

Armed with the Cayley graph construction, we can produce a vast wonderland of regular graphs, for the theory of finite groups provides us with a wide variety of possible starting points: $\mathbb{Z}_n$, dihedral groups, symmetric groups, alternating groups, matrix groups, direct products, semidirect products, wreath products, and so on. Moreover, the richness of group theory means that we can take advantage of the algebraic structure underlying a Cayley graph to extract information about

the graph. This theme permeates throughout this book, especially Chapter 2, Chapter 4, and Part III.

We remark that not every regular graph is a Cayley graph. The graph Y in Figure I.1, for example, is not a Cayley graph—see Exercise 10 of Chapter 2.

3. A TALE OF FOUR INVARIANTS

Computing the isoperimetric constant $h(X)$ of a graph X with n vertices requires minimizing over *all* sets with $\leq n/2$ vertices. The number of such sets grows exponentially as a function of n. Hence, generally speaking it is not feasible to determine $h(X)$ when n is large. So in our search for expander families (X_k), we seek indirect methods for showing that $h(X_k)$ is uniformly bounded away from 0. Typically, one looks for graph invariants that are related to the isoperimetric constant but more tractable to work with. We briefly describe two such invariants; the precise details of what they are and why and how they are related to the questions at hand occupies the remainder of this book.

Consider the complex vector space of functions from V to $\mathbb{C}$, where V is the vertex set of a graph X. Define a linear operator A from this vector space to itself, where the value of Af at a vertex v equals the sum of the values of f at all vertices adjacent to v. All eigenvalues of A are real, as we state in Def. 1.38. Denote by λ_1 the second largest eigenvalue of A. We shall see that λ_1 and $h(X)$ are intimately related.

In the special case where X is a Cayley graph, we obtain another important invariant by considering the underlying group G. We can decompose the operator A in terms of irreducible representations (irreps) of G. We define the *Kazhdan constant* κ relative to the group G and a subset Γ of G to be the minimum value ϵ such that for any nontrivial irrep ρ of G and any vector $\mathbf{v}$ in the representation space, some element $\gamma \in \Gamma$ moves $\mathbf{v}$, via ρ, a distance of at least ϵ.

Four invariants—the isoperimetric constant, the second largest eigenvalue, the diameter, and the Kazhdan constant—recur throughout this book. Each measures in some way the expansion quality of a Cayley graph. Table I.1 lists the four invariants, the notation used to denote them, the place they're defined, and some minor variations on them.

Table I.1 FOUR INVARIANTS

Invariant	**Notation**	**Defined in**	**Variation**
Isoperimetric constant	h	Def. 1.63	—
Second largest eigenvalue	λ_1	Def. 1.38	λ
Diameter	diam	Def. 1.18	—
Kazhdan constant	κ	Def. 8.5	$\hat{\kappa}$

We repeatedly find occasion to ask three questions pertaining to these four invariants.

1. How do they relate to one another?
2. How do they behave with respect to subgroups and quotients?
3. For large regular graphs with fixed degree, what are their best-case scenarios?

We provide the following answers.

(1) Table I.2 shows inequalities that relate each invariant to the others for a finite d-regular graph X.

(2) For quotients, we have the following. Let G be a finite group; $H \lhd G$; Γ a symmetric subset of G; $X = \mathrm{Cay}(G, \Gamma)$; and $Y = \mathrm{Cay}(G/H, \overline{\Gamma})$, where $\overline{\Gamma}$ is the image of Γ under the canonical homomorphism. Then:

$$h(X) \leq h(Y) \qquad \text{(Lemma 2.17)}$$

$$\lambda_1(X) \geq \lambda_1(Y) \qquad \text{(Prop. 2.26)}$$

$$\mathrm{diam}(X) \geq \mathrm{diam}(Y) \qquad \text{(Prop. 4.35)}$$

$$\kappa(G, \Gamma) \leq \kappa(G/H, \overline{\Gamma}) \qquad \text{(Prop. 8.28)}$$

For subgroups, we have the following. Let G be a finite group; $H < G$; Γ a symmetric subset of G; $d = |\Gamma|$; $\hat{\Gamma}$ a set of Schreier generators (see Definition 2.30); $X = \mathrm{Cay}(G, \Gamma)$; and $Z = \mathrm{Cay}(H, \hat{\Gamma})$. Then:

$$h(X) \leq \frac{h(Z)}{[G : H]} \qquad \text{(Lemma 2.41)}$$

$$\lambda_1(X) \geq \frac{\lambda_1(Z)}{[G : H]} \qquad \text{(Lemma 2.47)}$$

$$\mathrm{diam}(X) \geq \mathrm{diam}(Z) \qquad \text{(Prop. 4.35)}$$

$$\kappa(G, \Gamma) \leq d^{1/2}\kappa(H, \hat{\Gamma}) \qquad \text{(Prop. 8.30)}$$

Roughly speaking, the moral of the story in each case is that a group can do no better than its subgroups and quotients.

(3) As previously noted, d-regular expander families exist for all $d \geq 3$. In other words, the best-case scenario for h is better than $o(1)$. For a sequence of d-regular Cayley graphs, the three quantities h, $d - \lambda_1$, and κ have the property that if one of them goes to 0, then so do the other two. (This fact follows from the inequalities in Table I.2.) So the best-case growth rates for $d - \lambda_1$ and κ are also better than $o(1)$.

For λ, we can be more precise. The Alon-Boppona theorem asserts that for fixed d and arbitrary $\epsilon > 0$, if X is a sufficiently large d-regular graph, then $\lambda \geq 2\sqrt{d-1} - \epsilon$. Chapter 3 discusses this theorem in detail. *Ramanujan graphs* achieve the asymptotically optimal bound $\lambda \leq 2\sqrt{d-1}$. Section 6 and Note 4 of Chapter 7 deal with two interesting families of Ramanujan graphs.

The best-case growth rate for diameters is logarithmic as a function of the number of vertices of the graph (see Prop. 4.6). Expander families achieve this optimal growth rate. Chapter 4 deals with this fact and the severe restrictions it places

Table I.2 How the Four Invariants Relate to One Another

	λ_1 or λ	diam	κ or $\hat{\kappa}$
h	$\frac{d-\lambda_1(X)}{2} \le h(X) \le \sqrt{2d(d-\lambda_1(X))}$ (Prop. 1.84)	$\operatorname{diam}(X) \le \frac{2}{\log\left(1+\frac{h(X)}{d}\right)} \log(\lvert X\rvert)$ (Prop. 4.6)	$h \ge \frac{\kappa^2}{4d}$ (Prop. 8.17)
λ_1 or λ	—	$\operatorname{diam}(X) \le \lceil \log(n-1)/\log(d/\lvert\lambda_1(X)\rvert)\rceil$ (Note 6 from Chapter 4)	$\kappa \ge \sqrt{\frac{2(d-\lambda_1)}{d}}$ (Prop. 8.18)
diam	—	—	$\kappa(G,\Gamma) > \frac{\sqrt{2}}{\operatorname{diam}(\operatorname{Cay}(G,\Gamma))}$ (Exercise 6 of Chapter 8)

on the types of sequences of groups that can possibly yield expander families via the Cayley graph construction.

4. APPLICATIONS OF EXPANDER FAMILIES

The explosion of interest in expander families in recent decades stems in no small part because of the wealth of applications they enjoy, particularly in computer science. In this section we give a brief overview of some of these applications. This book deals only with the mathematical theory underlying expander families, but we want to briefly mention some applications so that the reader is aware they exist. Discussing them in detail is beyond the scope of this book. Instead, we point the reader to as many resources as we can.

4.1 Computer Science

There are many objects of interest to computer scientists that depend on expander families for their construction. We begin by quoting a paragraph from M. Klawe's article [79]. We have changed the citation numbers to match the ones in the back of this book.

> The study of the complexity of graphs with special connectivity properties originated in switching theory, motivated by problems of designing networks able to connect many disjoint sets of users, while only using a small number of switches. An example of this type of graph is a superconcentrator, which is an acyclic directed graph with n inputs and n outputs such that given any pair of subsets A and B of the same size, of inputs and outputs respectively, there exists a set of disjoint paths joining the inputs in A to the outputs in B. Some other examples are concentrators, nonblocking connectors and generalized connectors (see [38], [110]). There is a large body of work searching for optimal constructions of these graphs ([108], [17], [34], [105], [95], [110], [109]). So far all optimal explicit constructions depend on expanding graphs of some sort.

Along those lines, Lubotzky [87, p. 4] presents a result of Gabber and Galil [63] that shows how to construct superconcentrators using expanders. The survey article by Hoory, Linial, and Wigderson [70] discusses the construction of superconcentrators via expanders. Chung [38] discusses nonblocking networks and superconcentrators, along with constructions of these objects using expander families and Ramanujan graphs. She cites [38], [93], [108], and [109] for a history of expanders and how they apply to communication networks.

The introduction in Klawe [79] gives many applications of expanders to computer science. These include the following two results. Ajtai, Komlos, and Szemeredi [2] use expanders to describe a sorting network of size $O(n \log n)$ and depth $O(\log n)$. Their construction solved a problem on sorting networks that had been open for 20 years [27]. Erdös, Graham, and Szemeredi [54] use expanding graphs to construct sparse graphs with dense long paths. These types of graphs occur in the study of Boolean functions and fault-tolerant microelectronic chips. The survey article [70]

gives other applications of expanders to computer science, including applications to complexity theory.

Alon [5] uses finite geometries to construct expander families. These graphs enable him to improve previous results on a parallel sorting problem that arises in structural modeling. He also gives applications to Ramsey theory. Pippenger [111] applies expanders to sorting and selecting in rounds. Bellar, Goldreich, and Goldwasser [20] use expanders to study quantitative aspects of randomness in interactive proofs.

The publications here—especially [70], [79], and [87]—contain many more applications of expanders in computer science and elsewhere.

4.2 Random Walks

Let X be a regular graph. Suppose one begins at a vertex of X and at each step chooses a neighboring vertex uniformly at random (i.e., of equal probability) to move to next. Let $\pi_i(x)$ be the probability that one is at vertex x after i steps of this random process. If X is nonbipartite (see Def. 1.16) one can show that $\lim_{i\to\infty} \pi_i(x) = 1/n$, where n is the number of vertices in X. That is, after many steps the distribution is approximately uniform. Moreover, one can show that the rate at which this distribution tends to the uniform distribution is controlled by $\lambda_1(X)$. The convergence is rapid in expander families—in other words, one gets lost quickly when walking randomly in expanders. This leads to a number of applications that use walks in expander families. For more information on random walks in graphs see Diaconis's book [49], the survey article [70], and Terras's introductory book [129].

Charles, Lauter, and Goren [36] construct provably collision resistant hash functions that are used in cryptography. They construct these hash functions using walks on expander families in which finding cycles is hard. They use two different families of expander graphs: the family constructed by Lubotzky, Phillips, and Sarnak [89] and Pizer graphs [112]. Random walks on expander families can be used to reduce the error in probabilistic algorithms while trying to save on random bits used [70].

4.3 Error-Correcting Codes

We outline an application of expander families to error-correcting codes. See the survey article by Hoory, Linial, and Wigderson [70] for more information.

Suppose that two parties wish to communicate over a noisy channel. Let n be fixed and $C \subset \{0, 1\}^n$ be a code. Define the Hamming distance between x and y in C, denoted by $d_H(x, y)$, as the number of bits that need to be flipped to get from x to y. Define $\text{dist}(C) = \min_{\substack{x,y\in C \\ x\neq y}} d_H(x, y)$. One scheme to communicate over the noisy channel is to do the following. Transmit information using codewords from C. When a codeword is received, use the Hamming distance to see which codeword in C is closest to it. This will be taken as the codeword that was originally sent. One can show that if the number of bit-flips introduced in transmission is bounded by the greatest integer less than or equal to $(\text{dist}(C) - 1)/2$, then the above scheme is guaranteed to work.

Let $\text{rate}(C) = \log(|C|)/n$. A family of codes $C_n \subset \{0,1\}^n$ is asymptotically good if there are some fixed constants $r > 0$ and $\delta > 0$ such that $\text{dist}(C_n) > \delta n$ and $\text{rate}(C_n) > r$ for all n. Note that an asymptotically good family of codes gives a way to create codes that simultaneously ensure a lower bound on both distance and rate. Probabilistic arguments are able to establish the fact that such families exist. Amazingly, explicit constructions of expander graphs give explicit constructions of such codes. For more details see [70].

PART ONE

Basics

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1

Graph Eigenvalues and the Isoperimetric Constant

In this chapter, we discuss the two graph invariants that will occupy the bulk of our attention throughout this book: the isoperimetric constant, and the second-largest eigenvalue. Because we assume no prior knowledge of graph theory, we begin by giving several basic definitions and facts from that field.

If we know that the isoperimetric constant of a finite graph X is at least a, then it follows that for any set S containing no more than half the vertices of X, there are at least $a|S|$ edges that connect a vertex in S to a vertex not in S. Roughly speaking, the larger the isoperimetric constant is, the faster and more reliable the graph is as a communications network.

The eigenvalues of a finite graph are defined to be the eigenvalues of its adjacency operator. The isoperimetric constant of a finite regular graph X is closely related to its second largest eigenvalue $\lambda_1(X)$. In Section 7, we state the fundamental inequality (Prop. 1.84) that relates these two invariants.

In this chapter, we provide some techniques to estimate $\lambda_1(X)$. The most frequently used technique is a version of the Rayleigh-Ritz theorem from linear algebra. To make this book self-contained, we have included Appendix A. This appendix contains information on most of the topics from linear algebra that we will need: eigenvalues, eigenvectors, symmetric matrices, and so on. For further information on linear algebra, we refer the reader to the textbook by Friedberg, Insel, and Spence [56].

1. BASIC DEFINITIONS FROM GRAPH THEORY

Definition 1.1 A *multiset* is a collection of objects where objects may appear in the collection more than once. The number of times that a certain object appears in a multiset is called the *multiplicity* of that object. If S is a multiset, then $|S|$ is the number of elements in S.

Example 1.2
Consider the multiset $S = \{a, a, 4, a, -1, -1, x, 15\}$. The multiplicity of the element a is three, the multiplicity of the element -1 is two, and the elements 4, x, and 15 each have multiplicity one. The size of S is $|S| = 8$.

Remark 1.3
The notation for sets carries over for multisets. For example, we write $a \in S$ where S is from Example 1.2.

One can a multiset more rigorously as a function from a set U to $\{n \in \mathbb{Z} \mid n \geq 0\}$. For example, the multiset in Example 1.2 can be defined as the function f where $f(a) = 3, f(4) = 1, f(-1) = 2, f(x) = 1$, and $f(15) = 1$.

Definition 1.4 A *graph* is composed of a *vertex* set V and an *edge* multiset E. The vertex set V can be any set. The edge multiset E is a multiset whose elements are sets of the form $\{v, w\}$ or $\{v\}$ where v and w are distinct vertices. An edge of the form $\{v\}$ is called a *loop*. We sometimes denote V by V_X.

If $\{v, w\} \in E$, then we say that v and w are *adjacent* or *neighbors*. We also say that the edge $\{v, w\}$ is *incident* to v and w. If $\{v\}$ is in E, then we say that v is *adjacent* to itself. We also say that the loop $\{v\}$ is *incident* to v.

If $v \in V$, then the *degree* of v is the number of edges $e \in E$ such that v is incident to e. We write $\deg(v)$ to denote the degree of a vertex.

The *order* of a graph X, denoted by $|X|$, is the number of vertices in the graph.

Remark 1.5
Throughout this book we mainly deal with graphs that have finitely many vertices and edges. In fact, the only infinite graph in this book is the universal cover of a regular graph given in Def. 3.25.

We draw graphs as follows. For each vertex we draw a dot. If two vertices are adjacent, then we connect the corresponding vertices with a line (or curve) in our picture. If a vertex is adjacent to itself, then we draw a loop at that vertex. When drawing a graph, the apparent distance between two vertices does not matter. Also, it doesn't matter if edges appear to cross each other.

Example 1.6
Consider the graph in Figure 1.1. The vertex set is given by

$$V = \{v_1, v_2, v_3, v_4\}$$

and the edge set is given by

$$E = \{\{v_1\}, \{v_1, v_2\}, \{v_2, v_3\}, \{v_2, v_3\}, \{v_3, v_4\}\}.$$

The degrees of the vertices are

$$\deg(v_1) = 2, \quad \deg(v_2) = 3, \quad \deg(v_3) = 3, \quad \text{and } \deg(v_4) = 1.$$

The order of the graph is 4.